

Understanding real-world effectiveness of nirsevimab and the characteristics of breakthrough RSV infections: a study protocol of individual-participant-level data meta-analysis (the GEMINI study)

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ABSTRACT

Introduction The recent rollout of nirsevimab across multiple countries represents an important advancement in respiratory syncytial virus (RSV) prevention. However, real-world evidence on how its effectiveness varies with infant age and time since administration, and on the characteristics of breakthrough infections, remains limited.

Methods and analysis In this study, we aimed to conduct an international collaborative study to assess the real-world effectiveness of nirsevimab by monthly interval of chronological age and time since administration and to understand the characteristics of breakthrough infections. Both individual-participant-level data (IPD) and aggregated data will be collected from international collaborators identified by systematic review, existing collaborative network and academic conferences. The effectiveness of nirsevimab will be assessed by a test-negative study design, and a two-stage IPD meta-analysis will be conducted for data synthesis; a one-stage IPD meta-analysis will also be conducted among a subset of IPD data with sufficient sample size as an exploratory analysis. We will also describe the epidemiological, clinical and molecular features of breakthrough infections and compare them with infants and young children who were either unvaccinated and infected or vaccinated but not infected. The study will provide detailed insights into the dynamic effectiveness of nirsevimab and the characteristics of breakthrough cases.

Ethics and dissemination The study protocol has been approved by the Ethics Committee of Nanjing Medical University (2025–179). Ethical approval for the contributing studies is expected to be obtained independently by local investigators from their respective ethics committees. Findings will be presented at international conferences and submitted to peer-reviewed journals for publication.

BACKGROUND

Respiratory syncytial virus (RSV) is a leading cause of acute lower respiratory tract infection (ALRI) and a major contributor to infant hospitalisation and mortality worldwide. In 2019, RSV was estimated to cause approximately 12.9 million ALRI episodes,

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Nirsevimab prevents respiratory syncytial virus (RSV) in infants, but real-world effectiveness by infant age and time since administration, as well as breakthrough infection patterns, are not yet fully understood.

WHAT THIS STUDY ADDS

⇒ Pools global individual-level and aggregated-level data to estimate the dynamic effectiveness of nirsevimab and characterise profiles of breakthrough RSV infections in infants and young children.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ By clarifying the dynamic effectiveness of nirsevimab and the profiles of breakthrough cases, this study will help optimise nirsevimab implementation in practice.

2.2 million hospital admissions and 66 300 deaths among infants globally.¹ Nirsevimab is a long-acting monoclonal antibody that targets the prefusion conformation of the RSV F protein, thereby inhibiting viral fusion and preventing infection.² Nirsevimab received initial authorisation in the European Union in November 2022,³ becoming the first immunoprophylactic agent globally approved for the prevention of RSV in the general infant population. Since then, regulatory approval has expanded rapidly; as of June 2025, nirsevimab had been authorised in 54 countries, 18 of which have incorporated it into their national immunisation programmes. Recent data show high uptake in several countries, including Spain (90.1%), Luxembourg (83.7%), France (76.5%), and the United States (51.2%).⁴



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Although several studies reported the real-world effectiveness of nirsevimab,^{5–9} the reported effectiveness was for the entire study period (eg, season) rather than by finer time window, such as different months since administration. Given the potential waning of monoclonal antibody titres over time and the age-dependent variation in infants' susceptibility to RSV infection, the effectiveness of nirsevimab might vary dynamically according to both infant age and time since administration, which is not yet fully understood. Moreover, more evidence on the effectiveness of nirsevimab beyond the first 6 months since administration is needed, particularly in settings with year-round RSV circulation and among infants at high risk for severe RSV disease.

In addition to effectiveness, a few studies have documented the clinical or genotypic features of breakthrough RSV infections despite nirsevimab prophylaxis.^{10,11} However, the limited number of reported cases has precluded robust conclusions regarding the epidemiological or clinical patterns of these events, such as whether there were specific predictors for the breakthrough infections. To address these gaps, pooling real-world data from multiple studies could enable more granular estimates of nirsevimab effectiveness and inform our understanding of the characteristics of breakthrough cases. Here, we reported the study protocol of a global-level study on real-world effectiveness of nirsevimab and characteristics of nirsevimab breakthrough cases (the Global Effectiveness Meta-analysis of Individual-participant-level data (IPD) for Nirsevimab study, GEMINI).

Aim and objectives

In this study, we plan to collaborate with international partners to compile real-world data on the epidemiology of infants receiving nirsevimab, including both individual-level and aggregated-level data.

The primary aims are to assess the dynamic effectiveness of nirsevimab in infants, stratified by age and time since administration, and to characterise the epidemiological and clinical features of breakthrough RSV infections. The specific objectives are:

- ▶ To estimate the dynamic real-world effectiveness of nirsevimab in infants during their first RSV season, stratified by infant age (0–<30, 30–<60, ..., 330–<365 days) and by time since administration (0–<30, 30–<60, ..., 330–<365 days after administration).
- ▶ To characterise and compare epidemiological and clinical profiles of infants during their first RSV season across three groups¹: breakthrough cases (nirsevimab recipients testing positive for RSV)²; RSV-positive infants who did not receive nirsevimab and nirsevimab recipients testing negative for RSV.
- ▶ As an exploratory outcome, to further estimate the dynamic real-world effectiveness of nirsevimab in the second RSV season, stratified by child age (366–<390, 390–<420, ..., 690–<730 days) and by

time since administration (366–<390, 390–<420, ..., 690–<730 days after administration).

Definitions

Acute respiratory infection or acute lower respiratory infection (ARI/ALRI) episodes are defined as clinical episodes of acute respiratory infection (including both upper respiratory infection and lower respiratory infection) or acute lower respiratory infection occurring in outpatient or emergency, or primary care settings. Hospital admission is defined as episodes requiring inpatient hospital care. Severe outcome is defined as hospitalised cases with clinical severity indicated by admission to intensive care, use of supplemental oxygen, mechanical ventilation, prolonged length of stay (≥ 5 days, based on the reported median length of stay in a global systematic review)¹², or death. Breakthrough cases are defined as laboratory-confirmed RSV infections in infants who received nirsevimab ≥ 7 days prior to symptom onset, to account for the RSV incubation period and the time required to reach peak antibody concentrations.⁸

METHODS

Study design and eligibility criteria

Depending on the study design of the corresponding primary studies, we will evaluate real-world effectiveness of nirsevimab using either test-negative design, case-control design or cohort design. Infants will be eligible for inclusion if they: (a) are eligible for nirsevimab according to local immunisation strategy; (b) had a laboratory-confirmed RSV test result (either positive or negative, excluding single serology results such as IgG or IgM); and (c) had documented information on nirsevimab administration status (either receiving or not receiving nirsevimab) prior to illness onset; in settings where the maternal vaccine was administered, information on the status of maternal vaccination was also required. For settings where RSV maternal vaccine was implemented, information on whether mothers of these infants received RSV maternal vaccine was required. This protocol for the individual participant data meta-analysis has been registered with PROSPERO (CRD420251266487).

Data collection

Potential collaborators will be identified by one of the three approaches. The first is through relevant publications identified by a systematic literature search (details in online supplemental material 1) in three databases, Embase (Ovid), Global Health (Ovid) and MEDLINE (Ovid), as well as three preprint repositories, MedRxiv, SSRN and Research Square. The second is through our existing collaborative network, the Respiratory Viral Global Epidemiology Network (RSV GEN).^{13–15} The third is through academic conferences, such as the ReSViNET RSV Vaccine for the World. In particular, we will proactively identify collaborators based in non-high-income

countries and in settings with less distinct RSV seasons, which were under-represented in the available evidence of nirsevimab effectiveness and breakthrough infections.

For the systematic review, we conducted a preliminary search in March 2025 to identify all relevant publications from January 2022 onwards and will conduct two updates, one in early 2026 to cover all publications up to 2025, and the other one in early 2027 to cover publications in 2026. Published studies reporting results on the effectiveness or safety of nirsevimab in real-world settings (ie, excluding randomised controlled trials) among children under 2 years of age will be included. Two reviewers from the study team will conduct screening independently.

We will invite the identified collaborators to contribute individual-level and/or aggregated-level data on nirsevimab use and RSV-associated ARI or ALRI. To ensure consistency and completeness, a standardised data collection template listing all variables of interest will be provided to investigators (see online supplemental material 1).

Study-level characteristics to be collected include country, region, multi-centre study (yes/no), setting (urban, rural or mixed), study design (eg, cohort, case-control including test-negative), study period (YYYY/MM–YYYY/MM), hospital setting (outpatient, inpatient or emergency department), outcome definitions (ARI, bronchiolitis or ALRI), clinical specimen types for RSV testing (eg, nasopharyngeal swab), RSV testing methods (eg, PCR or antigen test), and data source for nirsevimab use (eg, self-reported or administration database / record). In addition, local immunisation strategies for nirsevimab and maternal vaccine will be documented for each participating region. For nirsevimab, this will include (i) whether a birth dose is offered during the RSV season (and, if so, the months considered as RSV season and the timing of the birth dose, ie, number of days after birth); (ii) whether, in addition to the birth dose, a seasonal catch-up administration is implemented (with details of the eligible age range, eg, 0 to <8 months); (iii) whether a year-round birth dose strategy is implemented; (iv) the timing of the birth dose administration (eg, within 24 hours of birth, 0–7 days after birth); and (v) the eligibility of high-risk children in their second year of life for receiving nirsevimab (including the definition of high risk, eg, age, comorbidities). For maternal vaccines, this will include: (i) whether maternal vaccine is approved; (ii) whether it is recommended; (iii) if recommended, whether the maternal vaccine or monoclonal antibody (mAb) is preferred; (iv) the gestational age (in weeks) eligible for maternal vaccination and (v) whether vaccination is administered seasonally or year-round.

Individual-level data variables will include anonymised patient identifiers (pseudo-ID), group identifier (when matching is used), sex, year and month of birth, receipt of nirsevimab (yes/no), year and month of administration, age of the child at nirsevimab administration (in days), dose of nirsevimab administration, birth weight (kg), gestational age (weeks+days), and method of gestational

age assessment (eg, ultrasound, last menstrual period). Where applicable, information on maternal RSV vaccination will also be collected, including maternal vaccine use (yes/no), year and month of maternal vaccine administration, age of child at maternal vaccine administration, and gestational week at maternal vaccination (in weeks). Clinical data will cover hospitalisation status and details (admission date, duration of stay), intensive care unit (ICU) admission, respiratory support (supplemental oxygen, non-invasive ventilation, invasive mechanical ventilation), RSV detection results and subtype, viral sequencing status, sequence database and accession number. Underlying medical conditions to be captured include congenital heart disease, immunocompromised status, Down syndrome, cystic fibrosis, bronchopulmonary dysplasia, pulmonary diseases and other specified comorbidities.

Aggregated data will be stratified by infant age in months at the time of healthcare presentation and by time elapsed since nirsevimab administration (or expected administration for those not receiving nirsevimab). Within each stratum, infants will be categorised by RSV test results (positive or negative), with separate counts for those who received and those who did not receive nirsevimab. These data are categorised into the three levels of severity: emergency or outpatient visits, hospitalisation and severe outcomes including intensive care unit admission, mechanical ventilation or death (based on the highest level of care).

Data management and ethics

Data sharing agreements will be established with contributing investigators as needed. The data will be transferred and stored using University of Edinburgh's secure service for data transfer, DataSync. Access is strictly limited to named members of the research team who have completed mandatory information security and governance training. The data will be destroyed by deleting the electronic records from DataSync (including from the recycle bin).

Ethical approval for the original studies is expected to be obtained by the contributing investigators from their respective local ethics committees. This protocol involves secondary analyses of de-identified data only. All investigators will be asked to confirm that data sharing aligns with their existing approvals. Ethical approval for the overarching protocol has been approved by the Ethics Committee of Nanjing Medical University (2025-179). Study data will not be used for any purposes beyond those specified in this protocol without the permission of the collaborators.

IPD integrity

Each dataset will be cleaned and checked for completeness by a statistician from the study team and cross-validated by a senior member of the team with extensive knowledge on the topic. Any discrepancies and

unreasonable values will be flagged out and clarification will be sought from the investigators. Both the original dataset and cleaned dataset will be maintained.

Quality assessment (Risk of Bias)

Study-level quality assessment will be conducted independently by two members of the research team. We will evaluate the risk of bias using a modified Newcastle-Ottawa Scale, considering the following domains: study design, case definition, data source for nirsevimab use, diagnostic testing method and confirmation of gestational age. The total score of this scale ranges from 0 to 6 points. Studies will be classified as low, moderate or high risk of bias, with thresholds of 5–6, 3–4 and 0–2 points, respectively. Detailed quality assessment criteria are provided in the Supplementary Material. Sensitivity analyses will be performed by excluding studies assessed as having moderate or high risk of bias, where applicable.

Statistical analysis

Effectiveness estimation

We will estimate the real-world effectiveness of nirsevimab using a test-negative design or a cohort design as appropriate, stratified by three outcome types¹: RSV-associated ARI/ALRI episodes,² RSV-associated ARI/ALRI hospital admission and³ RSV-associated ARI/ALRI with severe outcome.

For individual-level data, one-stage IPD meta-analysis will be applied; specifically, generalised linear mixed models (GLMMs) will be used to account for heterogeneity between studies. Logistic regression or Poisson regression will be applied to estimate ORs or relative risks (RRs), depending on the study design. Vaccine effectiveness (VE) will be calculated as $(1 - \text{OR}) \times 100\%$ or $(1 - \text{RR}) \times 100\%$, as appropriate. Models will adjust for potential confounders including infant sex, birth weight, gestational age grouping, nirsevimab dose, age at presentation, time since administration and geographic region.

For aggregated data, we will apply a two-stage IPD meta-analysis; for a test-negative study, within each age group and time-since-administration stratum, we will calculate the OR of prior nirsevimab exposure between RSV-positive and RSV-negative infants using 2×2 contingency tables; study-specific ORs will then be pooled using random-effects meta-analysis; likewise, we will calculate the RR for a cohort study and apply random-effects meta-analysis. Heterogeneity across studies will be assessed using the I^2 statistic and Cochran's Q-test.

Characterisation of breakthrough RSV infection cases

To characterise the epidemiological and clinical profiles of breakthrough RSV cases, we will compare three groups: (1) nirsevimab recipients testing positive for RSV (breakthrough cases), (2) RSV-positive

infants who did not receive nirsevimab, and (3) nirsevimab recipients testing negative for RSV.

Descriptive analysis will summarise continuous variables as means (SD) or medians (IQRs), depending on distribution, and categorical variables as counts and percentages. Group comparisons will be performed using χ^2 or Fisher's exact tests for categorical variables, and t-tests or Mann-Whitney U tests for continuous variables, as appropriate.

Multivariable logistic regression models will be fitted to identify factors independently associated with breakthrough RSV infection among nirsevimab recipients, adjusting for potential confounders such as age, sex, birth weight, gestational age, dose and timing of nirsevimab administration, and underlying medical conditions (eg, congenital heart disease and immunocompromised status). Disease severity indicators (hospitalisation, intensive care unit admission and need for respiratory support) will also be compared across groups using similar regression approaches. Same to the analysis of effectiveness, individual-level data will be analysed by one-stage model and aggregated data will be analysed by two-stage model.

Molecular characterisation of breakthrough RSV strains

We will compare viral genetic sequences from breakthrough cases (nirsevimab recipients testing RSV-positive) and RSV-positive infants who did not receive nirsevimab to identify potential genetic differences. Multiple sequence alignment will be performed, and phylogenetic trees will be constructed to assess the genetic relatedness of RSV strains. Mutations in the F protein, especially within the nirsevimab binding region, will be examined. Differences in mutation frequencies between groups will be tested using appropriate statistical methods (eg, Fisher's exact test). Where possible, associations between viral mutations and clinical severity will also be explored.

DISCUSSION

To our knowledge, this will be the first study to estimate the granular, real-world effectiveness of nirsevimab by infant age and time since administration. Unlike clinical trials, which are limited by fixed follow-up periods and narrow inclusion criteria, the use of real-world data allows for a more dynamic and generalisable assessment across diverse infant populations and healthcare settings.

By aggregating sporadic breakthrough RSV cases reported across individual studies and surveillance platforms, we expect to identify patterns of characteristics between breakthrough cases and non-breakthrough cases—an area for which published evidence remains scarce. Understanding the epidemiological, clinical and virological features of these cases may provide valuable insights into potential

mechanisms of immune failure and help inform the optimisation of RSV prevention strategies.

This study is expected to face challenges regarding heterogeneity across data sources, including variations in RSV diagnostic criteria, nirsevimab administration records and follow-up durations. These challenges may affect comparability and generalisability but will be mitigated by applying standardised data collection and harmonisation approaches as well as statistical models that adjust for heterogeneity, along with sensitivity analyses.

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